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AI-Augmented Research

Session 2.1: Validity Reasoning in AI-Mediated Data Work

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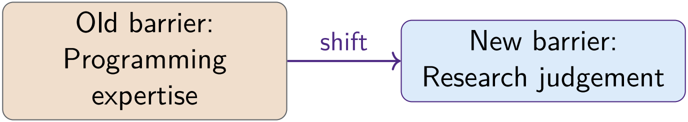
Session 2.1 Overview

Computational Research No Longer Requires Programming Expertise

A persistent assumption in academic research training is that computational methods require programming expertise. This assumption is **increasingly false**.

AI-mediated interfaces now allow researchers to perform data collection, cleaning, analysis, and visualization through natural-language interaction. The barrier to computational research has shifted:



Nature (2025): AI agents can handle complex, multi-step research processes.

What Computational Research Requires in an AI Environment

Requirement	**Description**	**AI can assist?**
Research question clarity	Specific enough to be operationalized	Partially (Tutor mode)
Data literacy	Understanding what data can support	No — researcher judgement
Validity reasoning	Assessing what to trust in AI outputs	No — researcher judgement
Verification discipline	Checking AI outputs against standards	No — researcher judgement
Documentation	Recording every step for reproducibility	Partially (Co-author mode)
Domain knowledge	Interpreting results in context	No — researcher judgement

AI-Assisted Data Collection: Opportunities and Limits

AI can assist with data collection by:

- Identifying relevant data sources
- Formulating structured queries
- Extracting information from unstructured text (PDFs, web pages, transcripts)
- Organizing extracted data into structured formats

Key accuracy benchmarks:

- Gartlehner et al. (2025, *Ann. Intern. Med.*): AI-assisted data extraction accuracy **91.0%**, recall **89.4%**, precision **98.9%** vs. human reference standard
- Peng et al. (2025): accuracy is **lower for image-based data** than textual data

> The 8–10% error rate means verification is non-negotiable for any AI-assisted data extraction step.

Structured versus Unstructured Data: Different Challenges

Data type	Examples	AI strengths	AI limitations	Verification
Structured	Surveys, databases	Cleaning, aggregation	Data type inference errors	Check summary statistics
Semi-structured	JSON, XML, PDFs	Extraction, normalization	Formatting errors	Spot-check extracted fields
Unstructured text	Transcripts, documents	Summarization, classification	Hallucination	Verify sample against source
Images/figures	Charts, scanned docs	OCR, description	Variable accuracy	Manual verification

Data Cleaning and Annotation: The Researcher's Role

Data cleaning and annotation are **methodological decisions**, not merely technical tasks. When an AI system cleans data — imputing missing values, standardizing categories, removing outliers — each decision reflects an assumption about the data-generating process.

The researcher must:

1. Understand and endorse each cleaning assumption
2. Document every cleaning decision and its rationale
3. Compare summary statistics before and after cleaning
4. Check that cleaning decisions are consistent with the analysis plan

Grimmer, Roberts & Stewart (2022, *Text as Data*): no value-free corpus construction; annotation is inseparable from theory.

Annotation Validity: AI as a Coder, Not an Authority

AI-assisted annotation can dramatically accelerate the coding of large text corpora. But AI is a **coder, not an authority**. Validation requirements:

1. Specify the annotation scheme in writing *before* running the AI
2. Code a random sample manually (minimum 10% or 100 items)
3. Calculate agreement between AI and human coding (Cohen's κ or equivalent)
4. Report the agreement statistic and the sample size in the methods section
5. Investigate and resolve systematic disagreements before using AI-coded data

$\kappa \geq 0.7$ is the minimum acceptable agreement for publication.

The Non-Determinism Problem: Reproducibility in AI-Mediated Analysis

LLMs are **structurally non-deterministic**: the same prompt to the same model can produce different outputs on different runs, even at temperature zero. This is not a bug; it is a consequence of parallel floating-point arithmetic in GPU inference.

- Ouyang et al. (2025, *ACM TOSE*): non-determinism undermines code consistency, developer trust, and research reproducibility
- Rover et al. (2026): non-determinism persists and **amplifies** in multi-turn interactions
- arXiv (2026): “background temperature” — hidden randomness in LLMs even at $T = 0$

AI-mediated analysis steps cannot be assumed to be reproducible without explicit documentation and re-run testing.

The Documentation Standard for AI-Mediated Computation

Zyphur (2026) Competency 2 specifies the minimum documentation standard:

1. Model name and version (e.g., GPT-4o, version 2025-01-01)
2. Temperature and any other generation parameters
3. Random seed where available
4. Date of the query (model behavior changes with updates)
5. **Re-run stability test:** run the same prompt 3–5 times; report whether outputs are substantively identical
6. Full prompt text (verbatim)
7. Full output (or representative sample for long outputs)

This is the minimum required for a reader to assess the reproducibility of the AI-mediated step.

Computational Reproducibility: New Challenges for AI-Mediated Research

AI-mediated analysis creates new challenges for the reproducibility norm:

1. **Model updates:** the same model name may refer to different underlying weights at different times
2. **Non-determinism:** even with identical parameters, outputs may vary
3. **Opaque training data:** the model's behavior may reflect undisclosed training data
4. **API deprecation:** the specific model version used may no longer be available

Baltes et al. (2025): non-determinism, opaque training data, and rapidly evolving models threaten reproducibility and replicability.

Peiris (2026, *Nature Astronomy*): the non-determinism of the LLM is no more scientifically relevant than the non-determinism of other stochastic tools, *provided the validated output is documented*.

Practical Strategies for Reproducibility

1. **Version locking:** use API calls with explicit model version parameters rather than "latest" aliases
2. **Output logging:** save all AI outputs to files with timestamps
3. **Re-run testing:** run the same prompt multiple times and report the variance in outputs
4. **Fallback documentation:** if the specific model version is no longer available, document the exact version used and provide the full prompt so readers can attempt replication with the closest available version

Validity Reasoning: The Core Competency for AI-Mediated Computation

Validity reasoning — the ability to assess what can and cannot be trusted in AI-mediated outputs — is the **core competency** for AI-mediated computational research. It requires:

1. Understanding the failure modes of the specific AI tool being used
2. Knowing which types of output are most prone to error (numerical calculations, citations, rare-domain knowledge)
3. Applying appropriate verification strategies for each output type
4. Reporting the verification steps and their results in the methods section

Zyphur (2026): Dimension 4 — AI-literate humans; validity reasoning as the foundation of responsible AI use.

The Six Observable Properties of Responsible AI Tools (Zyphur Dimension 3)

Property	Description	Why it matters
Verifiable citations	Source links resolve to real documents	Enables citation verification
Data residency controls	Researcher data not used for training	Protects participant confidentiality
Uncertainty reporting	Tool reports confidence or uncertainty	Enables calibrated interpretation
Reproducibility at known version	Specific model version is documented	Enables re-run testing
Auditability via session logs	Full interaction history is exportable	Enables methods documentation
No training on input	Inputs not incorporated into future weights	Protects intellectual property

Zyphur (2026), Dimension 3.

AI-Supported Coding Without Programming Expertise

AI coding assistants (GitHub Copilot, Claude, ChatGPT with code interpreter) can generate, explain, and debug code from natural-language descriptions. The verification discipline for AI-generated code:

1. Run the code and check whether the output is plausible
2. **Test with a small, known dataset** where the correct answer is known
3. Ask the AI to explain each line of code in plain language
4. Check the explanation against the output
5. Document the code, the prompt that generated it, and the verification steps

The test-with-known-data step is the most important: it is the only way to verify that the AI-generated code does what the researcher intends.

No-Code versus Low-Code versus AI-Assisted Coding

Level	Description	Examples	Verification discipline
No-code	Point-and-click, no code visible	Julius AI, Tableau, JASP	Check outputs against known benchmarks
Low-code	Minimal code, mostly configuration	Orange, KNIME, Weka	Review the workflow logic
AI-assisted coding	Natural language to code	ChatGPT Code Interpreter, Claude	Verify code logic; test with known data

Structured Failure-Mode Reporting (Zyphur Competency 6)

Zyphur (2026) Competency 6 requires that AI-mediated evaluations and syntheses report three elements together:

1. **Protocol:** the exact procedure used (model, version, prompt, parameters)
2. **Threat model:** the specific failure modes that could affect this type of output (hallucination, non-determinism, sycophancy, coverage gaps)
3. **Failure-discovery rate:** the proportion of AI outputs that failed verification, and how failures were handled

A methods section that reports a 5% failure rate in AI-assisted annotation, with a description of how failures were resolved, is **more credible** than one that reports no failures.

Session 2.1 Summary

Three things to carry forward

1. **Computational research in an AI environment requires validity reasoning, not programming expertise:** the ability to direct, verify, and critically assess what AI-mediated tools produce
2. **Non-determinism is a structural property of LLMs:** documentation of model version, temperature, seed, and re-run stability is the minimum required for reproducibility
3. **Structured failure-mode reporting** (protocol + threat model + failure-discovery rate) is the standard for reporting AI-mediated computational steps

References

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Questions and Discussion

Thank you!

Questions?

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